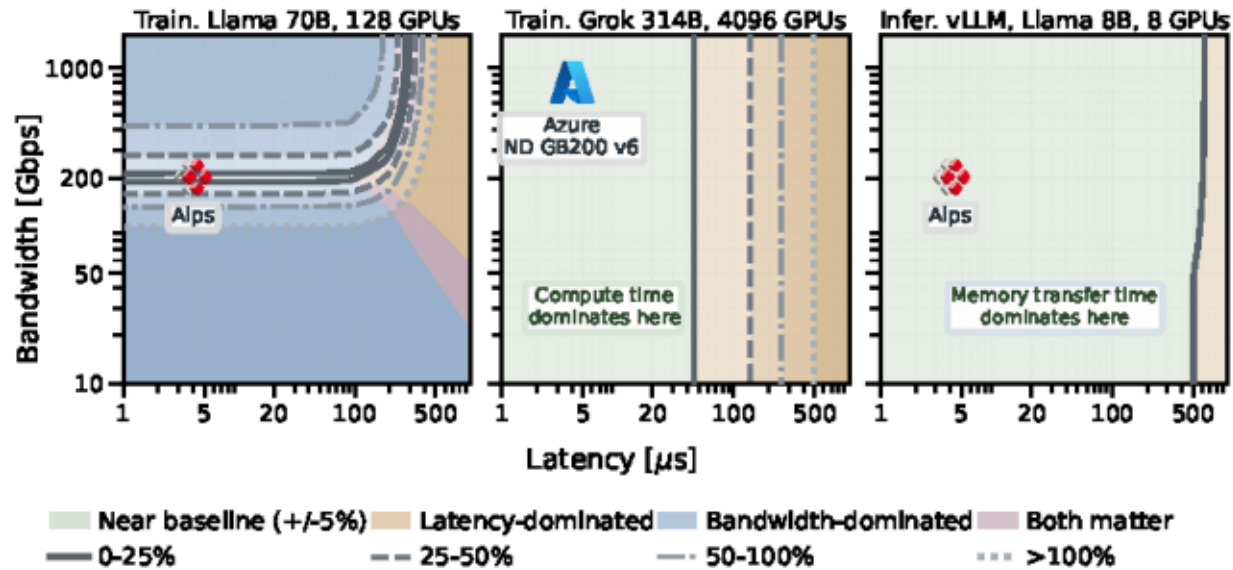


Figure 1: workload sensitivity maps

Paper plot



Reproduced plot from supplied numerical data

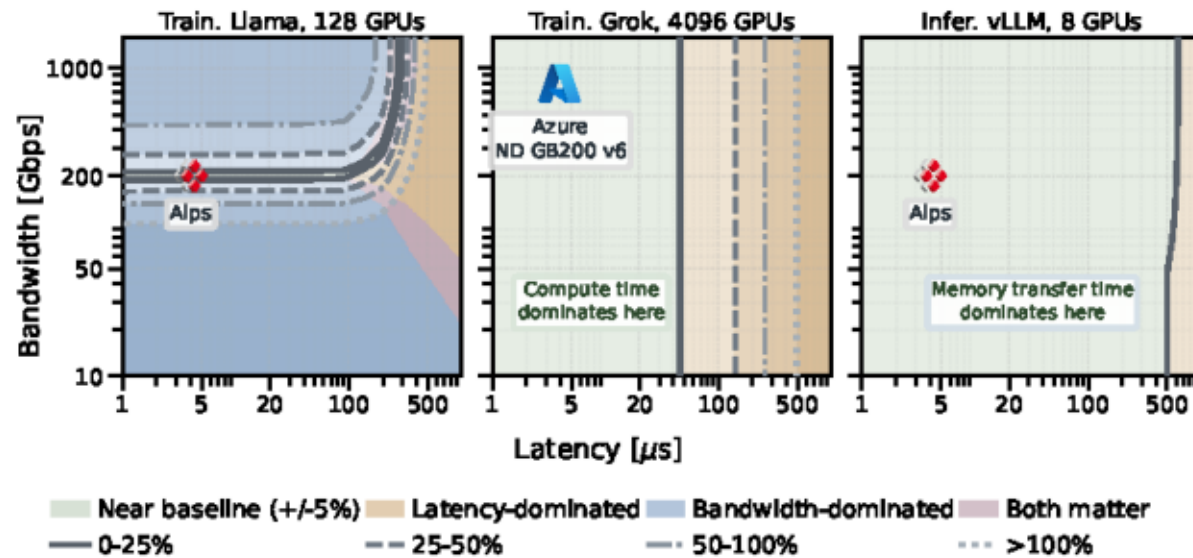
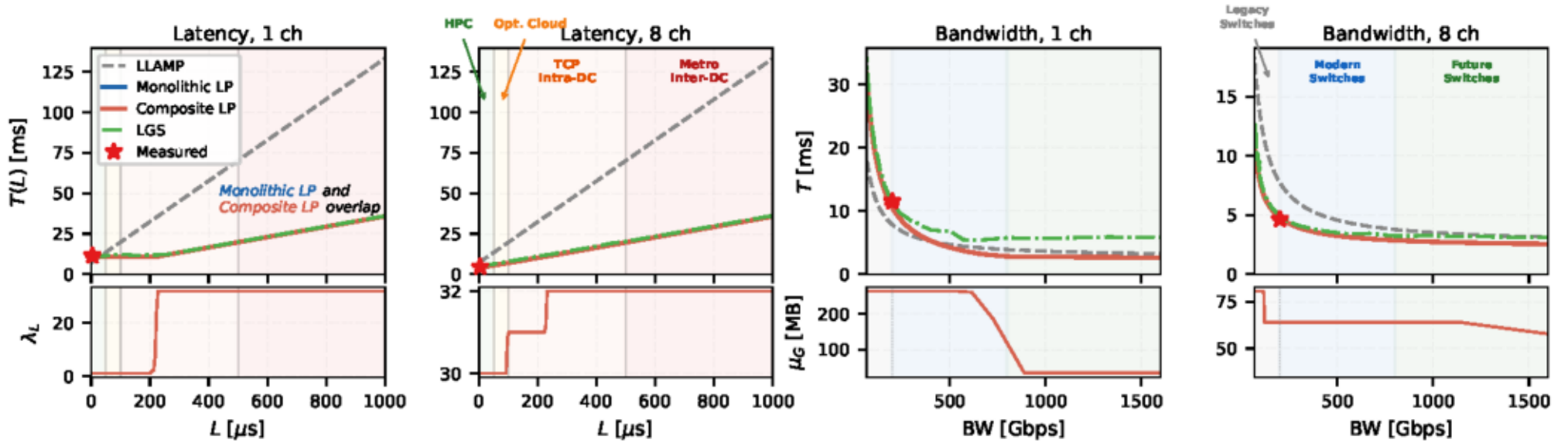
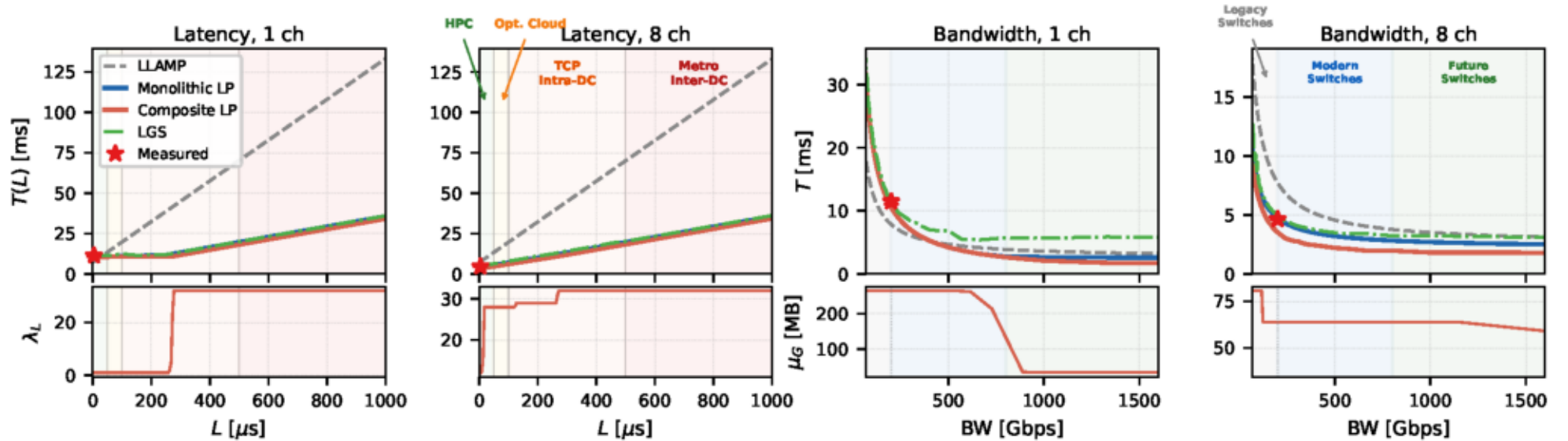


Figure 3: 128 MiB AllReduce

Paper plot



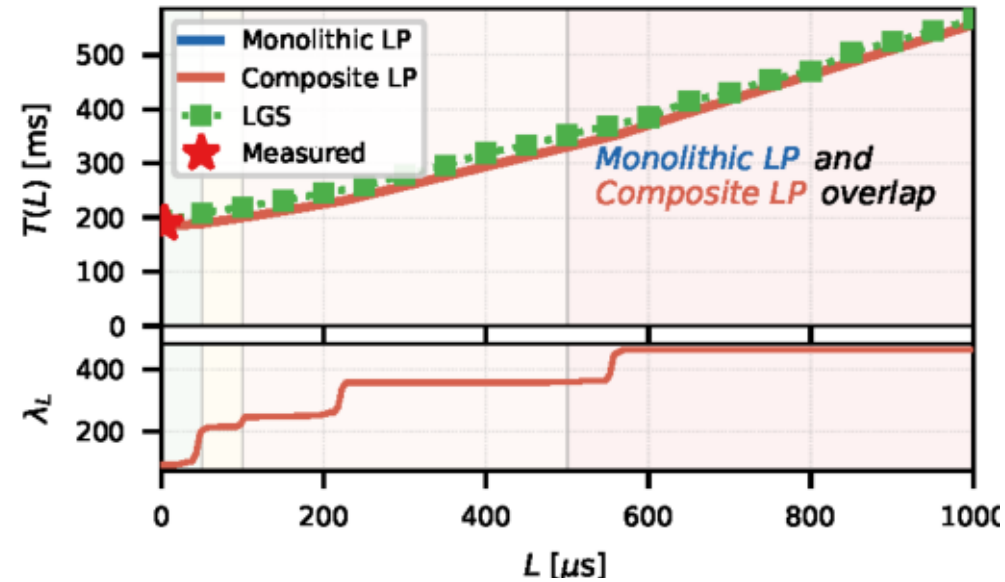
Full workflow output



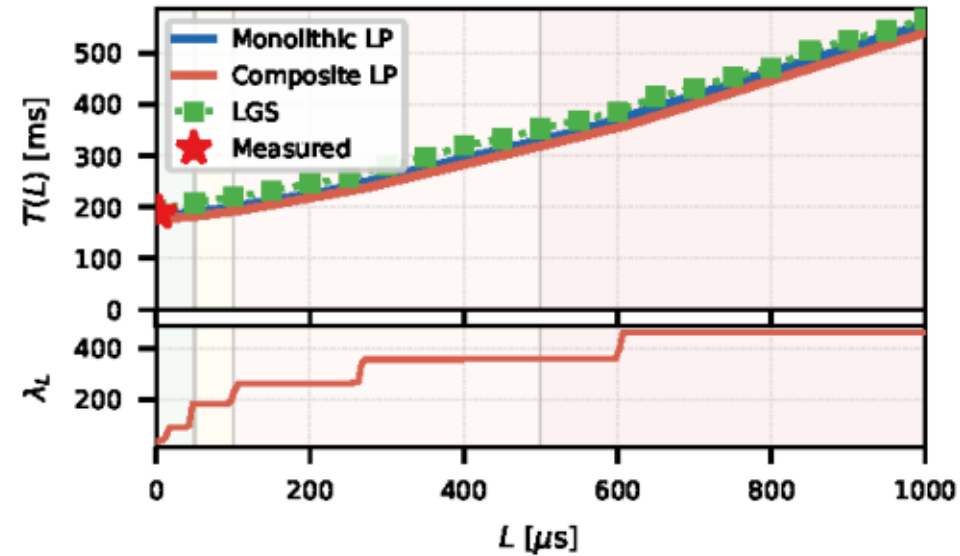
The Composite latency and bandwidth curves are regenerated by the full workflow. Raw trace mode also regenerates the Monolithic and LGS curves; LLAMP and measured inputs are supplied.

Figure 4: mixed collectives

Paper plot



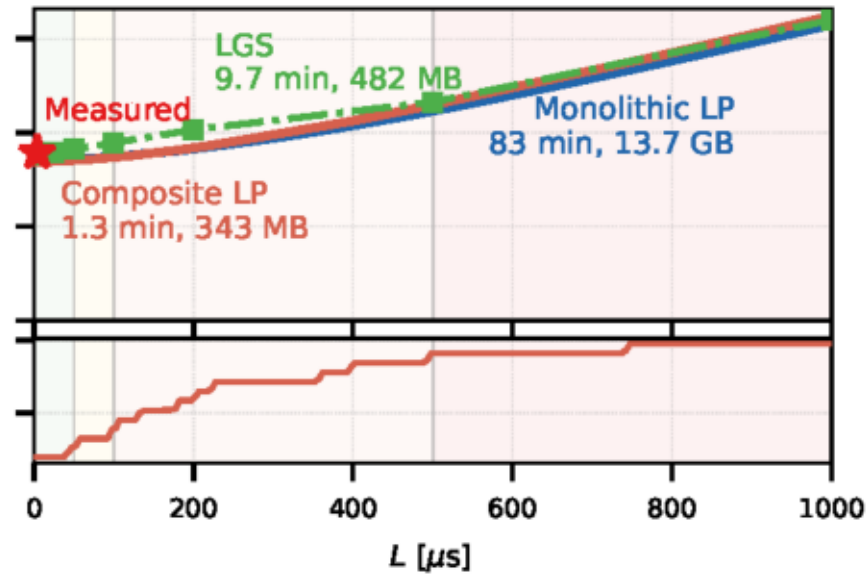
Full workflow output



The Composite curve is regenerated by the full analysis task. Raw trace mode also regenerates the Monolithic and LGS curves; the measured input is supplied.

Figure 5: Llama training iteration

Paper plot



Full workflow output

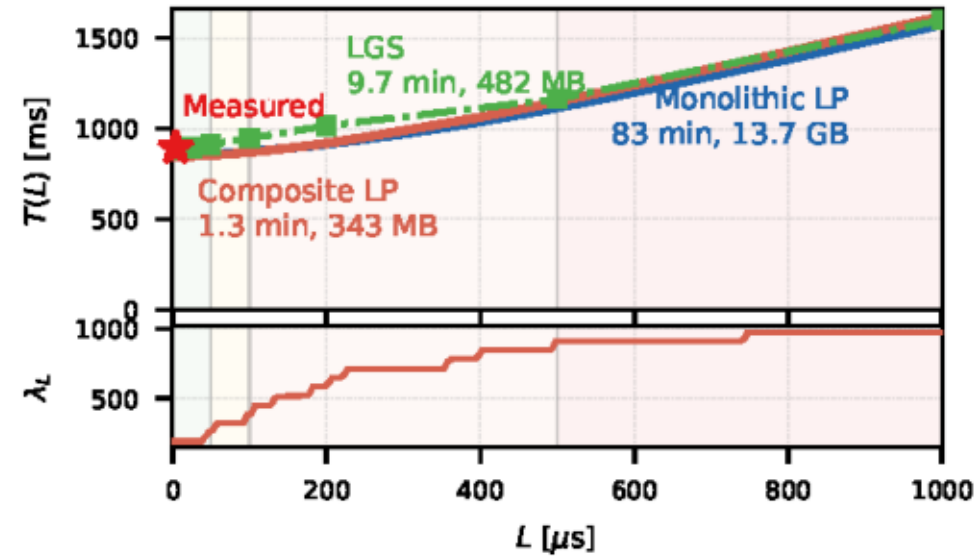
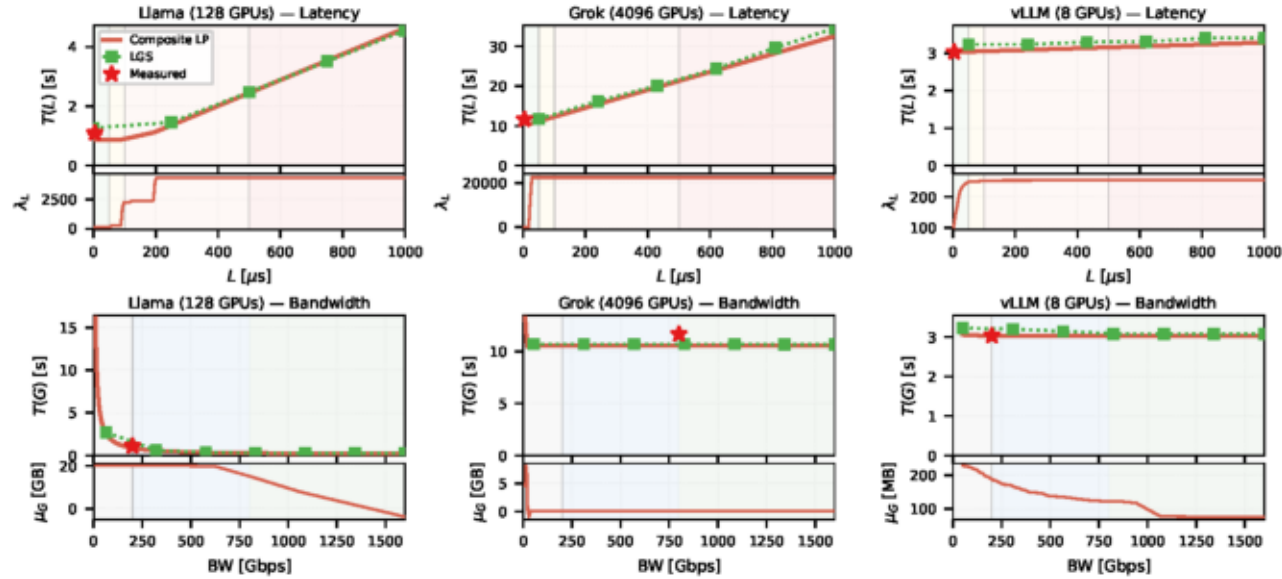
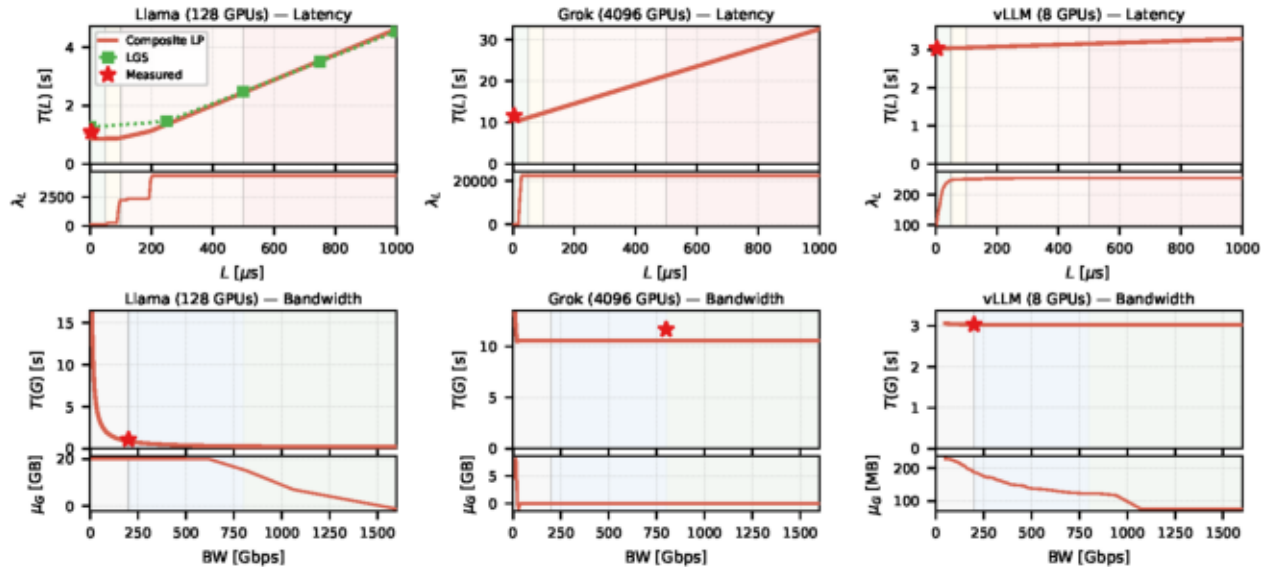


Figure 6: latency and bandwidth sensitivity

Paper plot

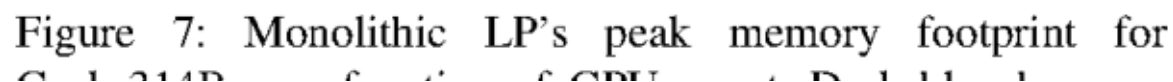


Full workflow output



The Llama latency and bandwidth curves are regenerated by the full workflow. Grok 4,096-GPU and vLLM use the supplied numerical inputs unless their optional analysis inputs are enabled.

Paper plot



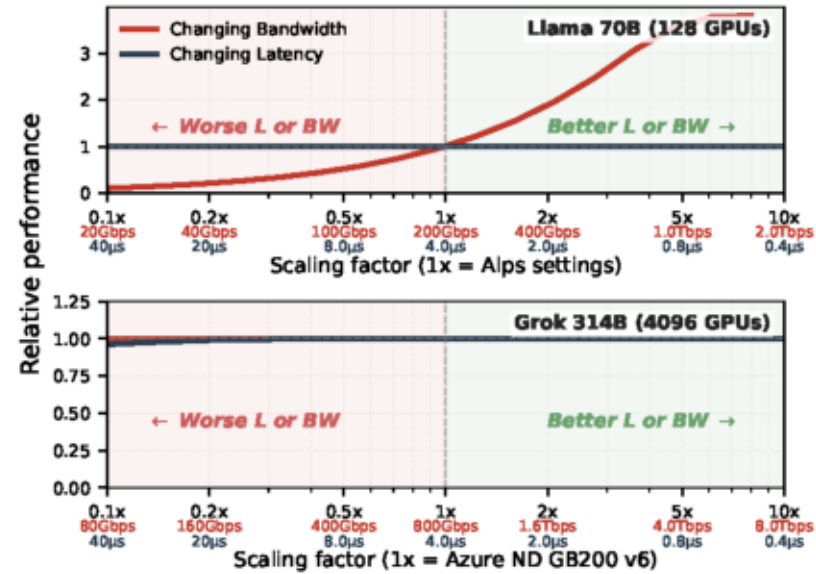
The chart displays Peak Memory [GB] on a logarithmic scale (from 10^1 to 10^5) against the Number of Nodes (N4, N8, N16, N32, N64, N128, N256, N512, N1024). The legend indicates that dark blue bars represent 'Measured' data and light blue bars represent 'Extrapolated' data. A red dashed line with circular markers shows the trend of memory usage, which increases exponentially with the number of nodes. A callout box highlights that at N1024, Monolithic LP needs ~34 TB, while Composition LP needs ~0.2 TB, which is 170 times less memory.

Number of Nodes	Peak Memory [GB]	Type
N4 (16)	8 GB	Measured
N8 (32)	10 GB	Measured
N16 (64)	22 GB	Measured
N32 (128)	59 GB	Measured
N64 (256)	146 GB	Measured
N128 (512)	389 GB	Measured
N256 (1024)	1.7 TB	Extrapolated
N512 (2048)	7.6 TB	Extrapolated
N1024 (4096)	34.0 TB	Extrapolated

At N1024: Monolithic LP needs ~34 TB vs Composition LP ~0.2 TB (170 times less memory)

Figure 8: network-parameter sensitivity

Paper plot



Reproduced plot from supplied numerical data

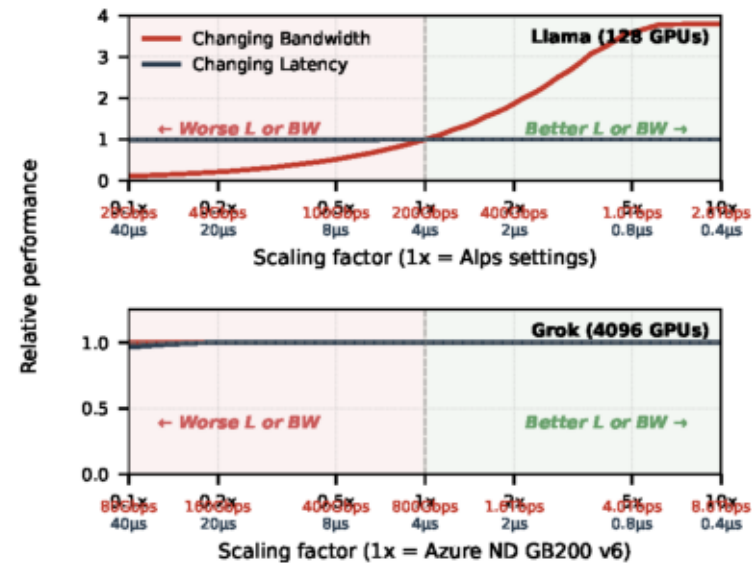
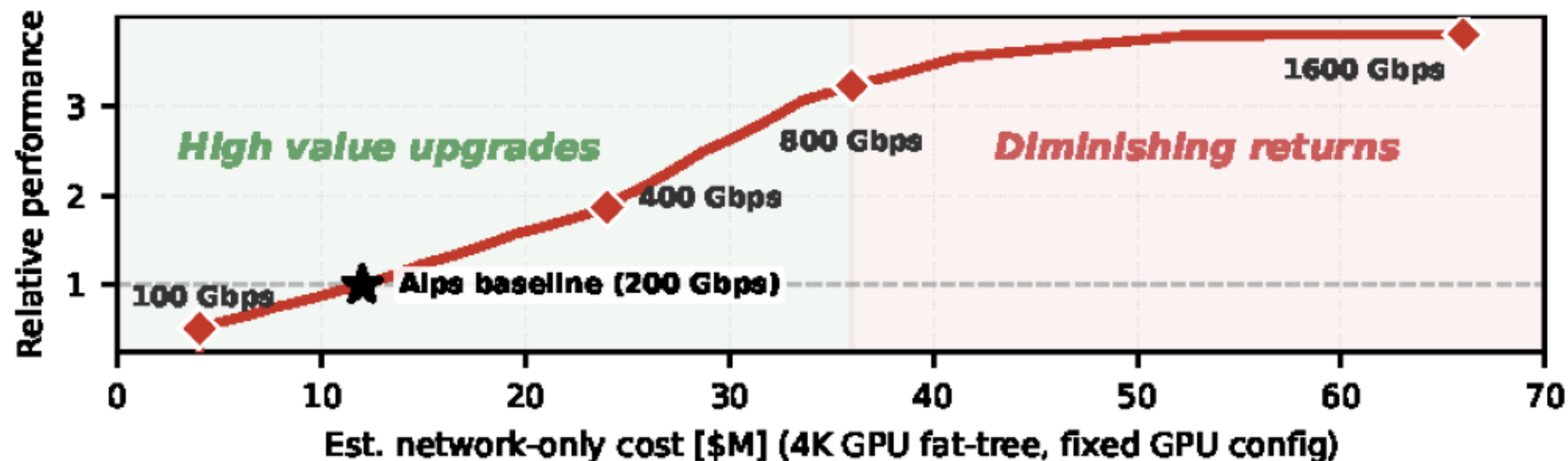


Figure 9: network cost trajectory

Paper plot



Reproduced plot from supplied numerical data

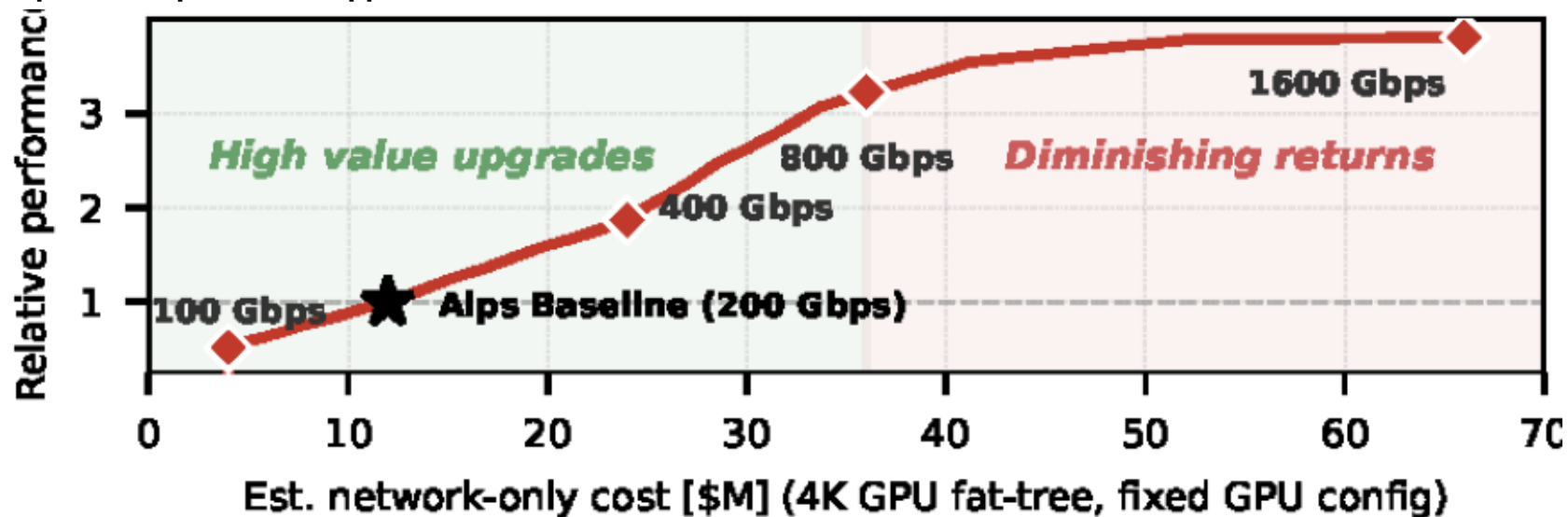
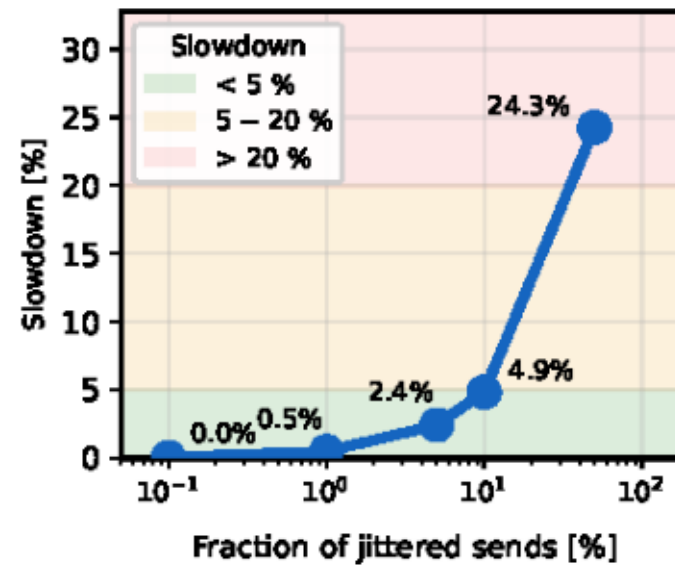
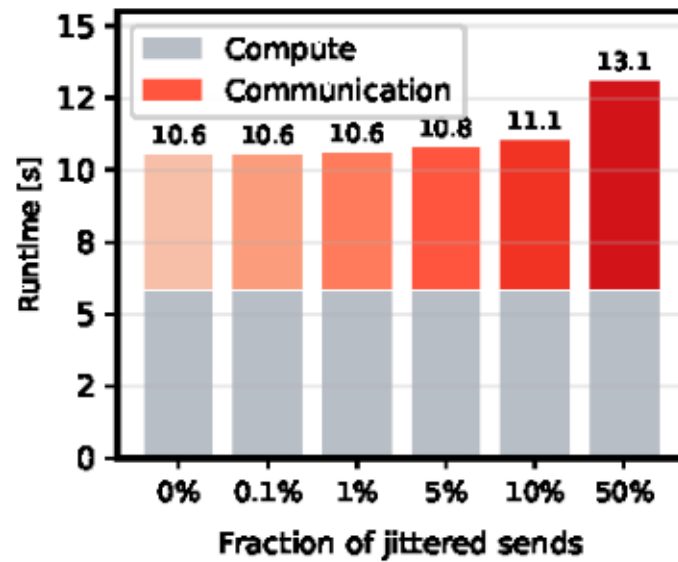


Figure 10: jitter sensitivity

Paper plot



Reproduced plot from supplied numerical data

